

ASHENFALL — A Soulslike RPG Pilgrimage

Design Document · v0.7 (REWORKED · magic promoted to core)
Minecraft 1.21.1 · NeoForge · Solo · Soulslike difficulty · Vanilla combat instincts preserved · Infinite world, landmark-driven · USW-style story world
Prepared 2026-09-26 · Supersedes v0.3 · v0.7 adds the magic layer

Companion documents

File	What it covers
DESIGN.md <i>(this file)</i>	The master design: pillars, combat, difficulty, world, story, bosses, gear, performance
MODS-INTERACTIONS.md	Full mod compatibility matrix — hard conflicts, soft conflicts, synergies, dependency chain, the unofficial-port risk register, and a 17-step build-time verification checklist
PROGRESSION.md	Tier-by-tier progression: entry requirements, gear and skill targets, landmarks, bosses, unlocks, skill curve, flask/waystone progression, boss prep checklists, collections, grinding loops, and a "I'm stuck" guide

<u>LORE.md</u>	The story bible: timeline, the Ember Seals, the nine nations and their contradictory accounts of the fall, key characters, all three acts, the 12 Annals, the 24 journal pages, boss lore, and the three endings
<u>RECOMMENDATIONS.md</u>	What to add to make it better — including a full Minecraft Dungeons element map, prioritised T1/T2/T3 additions, anti-recommendations, and accessibility toggles
<u>PHASES.md</u>	The build phases M0–M9 (plus M3b/M3c): what each phase adds, its verification gate, its locks, and the cross-phase "don't forget" registers
<u>PIPELINE.md</u>	The implementation pipeline: repository layout, mod management and version pinning, the worldgen order, the progression gating chain, the story delivery chain, the optimization budget and measurement protocol, and a 24-item oversight register
<u>DUNGEON-MODS.md</u>	Every mod that adds dungeon-like content, in five categories, with what each is for and the dungeon-content budget
<u>NARRATIVE-SYSTEMS.md</u>	How dialogue, rumours, boss monologues and lore are actually implemented — the dialogue engine, the JSON layout, NPC templates, the rumour register, cutscene code, resource packs, and the verdict on custom mods
<u>README.md</u>	How to use the pack builder (<code>tools/build.py</code>) and the workspace space model

✓ **Approved and folded in:** the T1 batch (Mini-boss Boss Bars, Configurable Boss Bars, Floating Damage Indicators, Loot Beams: Refork, Mob Champions, Boss Music Mod, YUNG's Traveler's Titles, the Rune/secret-mission system and the Ashenfall Threat tiers) is now in the stack — see §6.4, §6.8, §6.10a and §8.3. The **Monsters & Spellbooks** (Necro school) addon is in as well.

Still open: R1–R6 in RECOMMENDATIONS.md, and D2/D6/D7/D8/D11/D12/D13 in §15.

0. What changed in v0.3 (and why)

Change	Reason
Better Combat REMOVED. Epic Fight stays out. Combat is now vanilla.	Both overhauls replace your instincts. You're good at PvP — the pack must build on that, not overwrite it.
Movement is the combat layer instead	ParCool (parkour) + Combat Roll (dodge) + the 1.21.11 spear backported. Depth comes from <i>where you stand and how you move</i> , not from a new move-set.
Spear dashing from 1.21.11 — YES	Backported Spears (NeoForge 1.21.1) backports the Mounts of Mayhem spear: jab, speed-scaled charge attack, and the Lunge enchantment that rockets you forward.
Attribute swapping — YES	It's a real vanilla 1.21.1 behaviour (MC-28289). We ship a small custom NeoForge mod to automate it, plus the manual technique as fallback.
World is INFINITE again. No border, no bounded world.	"The beauty of Minecraft is that it's nearly infinite." Correct.

Replaced the bounded world with the Landmark Chain	Infinite world, but progression is gated by <i>finding</i> specific named landmarks — USW's "important areas you have to explore to progress."
New §8: Puzzles & Ancient Lore	USW is puzzle-and-lore driven. Riddle vaults, elemental gates, ordering puzzles, and lore that must be collected in order to unlock the ending.
Performance plan reworked	No pregen-the-world. Instead: starter-region pregen, on-demand pregen, entity caps, and a strict render-distance policy.
NEW: World fullness layer	Tree felling, real horses with gaits and tack, backpacks with upgrades, guarded villages, cooking. So the world feels <i>lived-in</i> , not generated.
NEW: Water & sailing layer	Real rivers carved into terrain, water that flows downhill with pressure physics, ocean currents/tides/swells, boats carried by currents. See §6.11.
NEW: §11.1 Where gear comes from	Direct answer to "villagers are disabled — so how do I get gear?" Ten sources, and why it is the point.
MAGIC PROMOTED TO CORE	Iron's Spells 'n Spellbooks is no longer phase 2 — it is a full parallel track with its own gear, bosses, workstations and faction politics. The Hexer class is live, the Arcane tier joins the ladder, the Eldritch school is lore-gated, and signature weapons gain imbued-spell slots.
NECRO SCHOOL APPROVED	Monsters & Spellbooks is in, not optional — the Necro school is the tenth school, older than the Empire. Highest-risk item in the stack; first to cut

T1 FEEDBACK BATCH APPROVED	Mini-boss Boss Bars, Configurable Boss Bars, Floating Damage Indicators, Loot Beams: Refork, Mob Champions, Boss Music Mod, YUNG's Traveler's Titles — all folded into §6
NEW: §8.3 The Ember Runes	Nine hidden runes across nine nation capitals open the Hidden Cathedral — a secret, optional endgame dungeon and the source of the Ashen tier's core material
NEW: The Ashenfall Threat	Seven per-region threat tiers raised by clearing that region's bosses at the current tier — the answer to keeping an infinite world dangerous
NEW: §5.5 The Dungeon Layer	Every vanilla structure reworked and renamed, Repurposed Structures' out-of-place ruins explained in-fiction as "displacement scars", The Graveyard's Lich side-quest, renewable dungeons, and the Lore Codex system that gives every ruin a name, a story, unique loot and a secret. When Dungeons Arise cut.

1. The pitch in one paragraph

Ashenfall turns Minecraft into a single-player action-RPG: you are a pilgrim crossing the **Known Lands of Vantyra** — an infinite world shattered into **nine nations** a thousand years after an empire fell in one night. You pick a class, you level real skills by using them, and you travel on foot and by waystone, following clues, compasses, and rumours to the **landmarks** that gate the story. Combat is **pure vanilla** — your PvP instincts are the foundation — amplified by parkour, dodge rolls, and a spear whose charge attack scales with how fast you're moving. Death is a mechanic: you drop your Embers where you fell and either recover them or lose them. There is a three-act plot, factions that remember what you did, ancient puzzles sealed inside empire ruins, and an ending that can only be reached by finding every piece of the story.

Not a kitchen-sink tech pack. **Not** a build pack. **Not** fair-by-default. **Not** a move-set overhaul.

2. Design pillars

#	Pillar	What it means in practice
1	Infinite world, curated highlights	The world never ends. But the story lives in ~20 named landmarks you must discover to progress.
2	Vanilla combat, movement-first	No combat rework. Your PvP instincts are the skill floor; parkour, rolls and spears are the skill ceiling.
3	Death is a mechanic	Ember loss, one-shot recovery run, finite healing, no keepInventory.
4	Gear gates content	You can't usefully enter a region below its tier. Every boss is a check on your build.
5	Bosses are story	Each boss is a faction champion, a seal-bearer, or a piece of the empire's ruin.
6	The land remembers	Faction standing changes what you can buy, who fights beside you, and which ending you can reach.
7	Lore is a mechanic	Ancient texts aren't flavour — they're keys, coordinates, and puzzle answers.
8	The world feels lived-in	Trees topple when you cut them, horses are real animals with gaits and tack, villages have guards worth defending, food is cooked rather than eaten raw, and your pack has weight and purpose.

3. Combat — vanilla instincts, movement depth

3.1 The rule: nothing replaces your hands

- **No Epic Fight.** Replaces vanilla move-sets per weapon; you re-learn combat from zero.
- **No Better Combat.** Rewrites attack chains, reach and timing; your PvP timing gets re-tuned under you.
- **What we keep:** vanilla attack cooldowns, crits, knockback, sweep, shield blocking, axe shield-break, mace smash & Breach, bow/charge mechanics, rod and pearl play, strafe-staggering, W-tapping, everything you already do well.

3.2 What we add (layers, never replacements)

Mod	What it adds	Why it doesn't fight your instincts
Backported Spears	The 1.21.11 spear: jab (fast, knockback, min-reach), charge (hold — damage scales with <i>relative speed</i> of you and the target; falling, minecart, elytra and mounts all multiply it; can dismount riders), and the Lunge enchantment (rockets you forward horizontally, costs durability + hunger, needs ≥ 6 food)	A <i>new weapon</i> , not a new way to use old ones. Minimum reach means you must manage distance — a skill you already have from PvP

Combat Roll	Dodge roll with i-frames, attributes and enchantments	A dedicated button for a thing PvP players already do with strafes and pearls
ParCool	Fast run, side dash, fence vault, flips, ledge grab, cliff cling, wall climb, crawl, fall-roll, wall run + its own stamina bar	Pure traversal. Never changes how a weapon feels in your hand
ParCool+ Compatibility+ +	Fixes animation clashes between ParCool / Player Animator	Stability

Combat tuning targets: roll i-frames in the first ~40% of the animation, 1 chained roll, ~1.1 s recharge; parkour stamina costs ~60% of the bar for a full escape route; bosses punish standing still and trading hits. Vanilla shield-timing stays a real skill.

3.3 Attribute swapping — how we get it

Attribute swapping is a genuine **vanilla 1.21.1 behaviour** (bug **MC-28289**): if you switch hotbar slots in the *same tick* as your attack packet, the server reads damage/attributes from the newly-selected item while using the cooldown you already charged on the old one. In practice: sword attack speed + mace damage/Breach in one hit.

- **Manual (always available, zero mods):** attack with the sword, swap to the mace on the same tick, swap back. Works in vanilla 1.21.1.
- **Automated (we build it):** a small custom NeoForge mod (~100 lines) that performs the ghost-swap on every critical hit — sword → mace → sword inside one tick — toggled with a key (default **G**), zero visual desync. The Fabric equivalent (`ghostswap` / Auto-Attribute-Swap) is tested on 1.21–1.21.4 but **has no NeoForge port**, so we ship our own.

- **Note:** Mojang has officially classified attribute swapping as an exploit and intends to patch it in future versions. Irrelevant for us — the pack is pinned to 1.21.1, single-player, forever.

3.4 Gear depth without a combat overhaul

Since weapon *feel* is vanilla, RPG depth comes from stats and mods that tune them:

Mod	Role
Attribute Modify (NeoForge 1.21.1)	Datapack-driven modification of item attributes — vanilla and Curios — so every modded weapon is tuned to sane, vanilla-consistent numbers
Reforged (fork of Tiered)	Every tool/weapon/armour you craft gets a tier + modifier with stat bonuses and abilities; stats can be rerolled at an anvil with a hammer (three hammer tiers, sold by villagers). Fully data-driven
Project MMO	Skill levels that grant real attribute effects

This is where "weapons to craft" gets its teeth — without ever changing how a swing feels.

4. Difficulty model — the "Soulslike Ruleset"

Ten hard rules.

1. **No natural regeneration.** You heal with **Ember Flasks** (Estus-style, finite charges).
2. **Flask charges refill only at rest points** — Waystones, boss arenas, taverns/camps. Dying does not refill them.
3. **Embers.** All XP is "Embers". On death you drop them at the death spot. Recover them, or die again en route and lose them forever.
4. **Hollowing.** Die while carrying un-recovered Embers → *Withering*

- (–10% max health, stacking to a –20% cap) until you recover your Ember pile.
5. **Stamina governs combat.** ParCool stamina + Combat Roll costs. Empty stamina = no dodge = death.
 6. **You dodge, you don't facetank.** Short i-frames; bosses punish trading hits.
 7. **No keepInventory, ever.** GraveStone Mod is a corpse locator only (grave marker, obituary, coordinates); item retrieval *at* the grave is disabled.
 8. **The world scales with distance.** The further you travel from where you woke, the stronger everything gets — the Ashenfall thickens outward. This is what makes an *infinite* world dangerous instead of empty.
 9. **Traders don't sell power.** No diamond/netherite gear, enchanted books, or potions from villager trades. Villages are lore, supply, and waystone hubs. (*Gear comes from crafting, dungeons, bosses, mobs, enchanting, bartering and faction standing — see §11.1.*)
 10. **One life is a run, not a save.** Optional extreme death rules ship **disabled** as a "Masochist" profile.

Difficulty curve target (solo)

Stage	Player HP	Boss HP	Deaths expected per boss (first attempt)
Overworld / first boss	20	150–250	3–8
Nether gauntlet	30–40	400–700	5–12
Aether	40	500–800	4–10
Hollow Court (Twilight)	40–50	800–1,500	6–15
Deep / Otherside	50	600–1,000	5–12
End / endgame gauntlet	60+	1,500–4,000	8–20

5. The world — infinite, but never empty

5.1 The model

No world border. No bounded world. No "pre-generate everything."

Instead, the world is infinite and layered by *danger* and *story density*:

Band	Distance from origin	What's there
The Heartland	0 – ~1,500	Norman Remnant and Spice Kingdom villages, starter dungeons, the first landmark. Safe-ish. Tuned for iron gear.
The Reaches	~1,500 – ~6,000	The other nations, When Dungeons Arise megastructures, the cold tower, the first real bosses. Tuned for diamond/netherite.
The Frontier	~6,000 – ~20,000	Rare biomes, Aether/Hollow Court portals hidden in remote ruins, the Otherside. Tuned for endgame gear.
The Ashen Wastes	~20,000+	Scaling is brutal, structures thin, nothing story-critical. For pilgrims who want to see how far the blight goes.

This is the USW move without the island: an infinite world where the story is concentrated in discoverable highlights, and the wilderness gets genuinely worse the further you push. The distance-scaling mods make "wander forever" a real risk/reward decision instead of a chore.

5.2 The Landmark Chain (the progression spine)

Progression is gated by **finding** landmarks — not by reaching a tier

number. There are ~**20 landmarks**; each chapter needs 1–3 of them.

How you find them (four systems, layered):

- 1. **Explorer's Compass** (NeoForge 1.21.1) — craftable compass that lists *every registered structure* from every installed mod, with distance + heading. The "legal /locate." Blacklisted per-pack so it can't just hand you the Cathedral.
- 2. **Cartographer's Maps** — dungeon loot items that, when read, pin a *specific* landmark's coordinates to your Xaero's map. Found in taverns, nation villages, and boss chests.
- 3. **Rumours** — FTB Quests + KubeJS dialogue: villagers and nation NPCs give vague directional clues ("the tower where the stars fall, three days north of the frozen river"). Real exploration, real dead ends.
- 4. **The Pilgrim's Journal** — lore pages that double as treasure maps. Some landmarks are *only* findable through the journal.

Why this works: you can never be soft-locked (compasses exist), but you're never handed the answer either. Every landmark is a thing you *found*.

5.3 Landmark list — the highlights

#	Landmark	Gates	How you find it
L1	The Drowned Chapel	Ember Flask upgrade #2	Journal page 1
L2	The Cold Tower	Night Lich / Seal Shard #1	Soul stars visible at night; Inuit North rumours
L3	The Burning Arena	Ignis / Mechanical Fusion Anvil	Nether fortress-adjacent; Explorer's Compass
L4	The Sunken Spire (Aether)	Aether portal catalyst	Cartographer's map in a Norman vault

L5	The Hollow Gate	Twilight Forest entry	Choir lore + Twilight portal materials
L6	The Maze Minotaur's Labyrinth	Hollow Court progression	Magic Map mechanics + journal page 9
L7	The Otherside Rift	Deeper and Darker	Explorer's Compass (deep dark biome)
L7b	The Catacombs / Lost Citadel / Echo of Tyros	Iron's Spells bosses — Dead King, Keeper, Fire Boss	Arcane-essence rumours from wizard NPCs; Explorer's Compass
L7c	The Eldritch Vault	The Eldritch Manuscript, which gates an entire spell school	Codex + journal pages only — no compass can find it
L8	The Cathedral of Ash	Endgame / Maledictus	Only via the completed Annals (§8)
L9–L20	Nine nation capitals, the Ember Forges, the Seal Vaults, the Merchant's Vault...	Standing, gear, endings	Mixed: rumours, maps, journal, standing-gated
L★	The Hidden Cathedral (<i>secret — not on the chain</i>)	The Ashen tier's core material + a hidden ending	Only the nine Ember Runes , each hidden behind a secret in a nation capital. No compass, no map, no rumour points at it

5.4 Worldgen stack

Mod	Layer
Terralith v2.6.x	~100 new biomes — the biome backbone
Explorify v1.6.x	Vanilla-friendly structures that feel handcrafted — density without megastructure fatigue
Dungeons and Taverns	Smaller dungeons + taverns (taverns = lore + bonfires)
Towns and Towers	Village/outpost variety, waystone seeding
When Dungeons Arise (<i>cut — see §5.5</i>)	Superseded by YUNG's + Structory + The Graveyard + Hopo; kept only as an optional profile
YUNG's Better suite (core)	Dungeons · Strongholds · Mineshafts · Ocean Monuments · Nether Fortresses · Desert Temples · Jungle Temples · Witch Huts · End Island (+ YUNG's API, Bridges/Extras optional) — every vanilla structure reworked , see §5.5 Tier A
Repurposed Structures	Biome variants of vanilla structures — the "displacement scars" of the Ashenfall (§5.5)
Hopo Better Ruined Portals / Underwater Ruins	Reworked ruined portals (incl. piglin structures) and underwater ruins
The Graveyard	10+ lore structures + The Lich's Bargain side-quest line (§5.5 Tier B)
Structory	Ambient ruins, towers, cottages, graveyards with hidden lore
Dimensional Dungeons (<i>renewable</i>)	Buildable portal → generated dungeon dimension; grinding layer + Hollow Challenges
Loot Integrations (<i>optional</i>)	Pushes modded gear into modded structure chests

Tectonic <i>(optional, perf-gated)</i>	Continent/island terrain reshaping. Beautiful, but heavier — see D6
Nature's Compass / Explorer's Compass	The discovery layer
Regions Unexplored <i>(dropped by default)</i>	Cut for perf unless biomes feel thin

Layering order: Terralith (biomes) → Explorify / D&T / T&T / WDA (structures).

Structure layering order: YUNG's (replace vanilla) → Repurposed Structures (variants) → Hopo (portals/underwater) → Structory / Explorify / D&T (ambient density) → Terralith (themed set pieces). **AW2 removed** — it was the only mod here with no 1.21.1 build. The structure layer is now 24 mods deep and is at its practical limit; do not add another structure mod without cutting one.

5.5 The Dungeon Layer — a world of ruins and lore

The world outside the nine capitals is where most of your hours go. This layer exists to make sure that **every vanilla structure type has been reworked, renamed and given a story**, and that the wilderness between cities is dense with custom ruins worth walking to.

Three tiers:

- **Tier A — Reworked vanilla.** Every vanilla structure type is redesigned and given an in-world name, a lore entry, and unique loot. Nothing generates "as vanilla".

- **Tier B — Custom & lore-rich structures.** Modded structures with their own stories, including a full side-quest line.
- **Tier C — Renewable dungeons.** A generated dungeon dimension for the grinding layer and the Hollow Challenges.

Tier A — Reworked vanilla structures

Vanilla structure	Rework mod	In-world name	Lore hook	Unique loot
Monster dungeon (spawner room)	YUNG's Better Dungeons	<i>The Delver's End</i>	Imperial mining-guild outposts, first Codex page	Codex page, Ember shard
Stronghold	YUNG's Better Strongholds	<i>The Seal Vault</i>	Where the Nine Ember Seals were kept; the End portal room is the Vault's heart	Annals fragment, vault key
Mineshaft	YUNG's Better Mineshafts	<i>The Deep Delve</i>	Abandoned imperial mine; Ember Forge blueprints	Forge blueprint, dark materials
Desert temple	YUNG's Better Desert Temples	<i>The Seljuk Sand-Vault</i>	Seljuk Expanse tribute vaults	Seljuk culture gear
Jungle temple	YUNG's Better Jungle Temples	<i>The Mayan Sun-Vault</i>	The Sun-Eater cult of the Mayan Deep	Blood/alchemy gear

Witch hut	YUNG's Better Witch Huts	<i>The Hedge- Witch's Hut</i>	Potion lore; where the Ember- Warden path begins	Rare potion recipes
Ocean monument	YUNG's Better Ocean Monuments	<i>The Drowned Choir</i>	The Byzantine Choir's sunken cathedral, guarded still	Choir relic, prismarine gear
Nether fortress	YUNG's Better Nether Fortresses	<i>The Burning Bastion</i>	Witchbane Crusader ruins; the piglin war	Anti-magic gear
End city	YUNG's Better End Island	<i>The Emperor's Last City</i>	The Empire's fall; the Harbinger's origin	End gear, Annals fragment
Ruined portal	Hopo Better Ruined Portals	<i>The Broken Gate</i>	How the flawed Tenth Ember was carried between worlds	Banners, potions, mob heads
Underwater ruins	Hopo Better Underwater Ruins	<i>The Drowned Province</i>	A coastal nation the Ashenfall swallowed	Nautical loot

Woodland mansion	vanilla + pillager outposts + Repurposed Structures illager variants	<i>The Illager March</i>	Where the Nine Nations' enemies hold out	Totems, vindicator gear
Ancient city	vanilla, scaled	<i>The Warden's Vault</i>	Deepest lore; the Otherside connection	Swift Sneak, echo shards
Trail ruins	vanilla	<i>The Old Road</i>	The imperial road network — and where its waystones died	Pottery sherds that are Codex pages
Igloo / shipwreck / village	Towns and Towers + Guard Villagers + Millénaire	—	Supply, lore, and the standing system	—

Repurposed Structures — the displacement scars. This mod takes vanilla structures and builds *biome variants* of them: a stonebrick Nether fortress overgrown in a jungle (silverfish-infested, shulker-rich, with a portal back to the Overworld); strongholds out past 8,000 blocks in the End; mineshafts and bastions in biomes where they have no business being. **In-world explanation: these are displacement scars** — places where the Ashenfall tore structures out of their world and dropped them somewhere else on the night the Seals broke. Finding a Nether fortress in a jungle isn't a mod quirk; it's a wound. That single in-fiction justification covers every "weird" structure in the pack.

Tier B — Custom & lore-rich structures

Mod	What it adds	Its story role
The Graveyard	10+ structures: Haunted House, Small/Medium/Large Graveyard (the large ones have riddles and puzzles), Crypt (a trap-filled labyrinth), Ruins, Lich Prison (a floating island above the oceans), plus Ghoulings, the Nameless Hanged (rare artifacts), sarcophagi and coffins as storage, bone dagger, dark iron	A full side-quest line: "The Lich's Bargain." Collect three Ominous Bone Staff fragments from three different Ruins, take a bone dagger from an Acolyte, fill a Vial of Blood, find the floating Lich Prison, and free the lich. Perfect soulslike side-content with a real payoff
Structory	Ruins, fire towers, cottages, stables, graveyards, settlements, boats — with seasonal theme updates and hidden lore; Terralith-compatible	The ambient "lived-in wilderness" layer. Small, frequent, cheap, and full of one-line stories
Explorify	Vanilla-friendly structures that feel handcrafted rather than generated	Density without megastructure fatigue
Dungeons and Taverns	Smaller dungeons + taverns	Taverns = lore, bonfires, rumours

Ancient Warfare 2 structures	DROPPED — no 1.21.1 build (CurseForge project 224602 stops at 1.12.2, last update 2021). Its role as "The Empire's provinces" is carried instead by Towns and Towers + Structory + Repurposed Structures , with the province naming and lore done in KubeJS — see §5.5 note	
Terralith structures	Biome-appropriate ruins and set pieces	Fills the new biomes with reasons to stop
Iron's Spells structures	The Catacombs (Dead King), the Lost Citadel (Keeper), the Echo of Tyros (Fire Boss), plus the magical structures that spawn spellcasters and hold runestones	The scattered schools of magic. These are the Choir's outposts, sacked on the night the Seals broke — the reason spellbooks and runestones are loot rather than recipes. Each has a Codex entry and a unique school relic

Tier C — Renewable dungeons

Dimensional Dungeons — one buildable portal anywhere in the world opens into a procedurally generated dungeon dimension: rooms, traps, loot, and a return portal. Renewable, repeatable, and the perfect home for the grinding layer and the **12 Hollow Challenges**. Its Patchouli guidebook doubles as an in-world field manual. Lore framing: the portal opens into *the space between* — the Ashenfall's own territory — which is why the loot there is good and the danger never resets.

The Lore Codex — the system that makes every ruin feel

authored

The mechanism tying it all together. **Every structure type in the pack has five things:**

1. **An in-world name** (table above) — so "you found a desert temple" becomes "you found a Seljuk Sand-Vault".
2. **A Codex entry** — a Patchouli page describing who built it, why, and what happened to them. 24 entries total, matching the 24 journal pages.
3. **Unique loot** — 1–2 items found *only* in that structure type, injected into its loot tables via datapack.
4. **A secret** — a hidden room, a riddle, a buried cache, or a mini-boss. Every structure type has at least one.
5. **Sometimes a hook** — a Cartographer's Map, a Codex page that is also a treasure map, or a rumour pointing at the next landmark.

Collection reward: completing a Codex set (e.g. all nine nation structures) grants a permanent upgrade and unlocks the next landmark's location. This is what turns "I found another dungeon" into "I'm reconstructing a dead empire".

Implementation: vanilla/modded loot tables edited by datapack, Patchouli Codex book, KubeJS for the riddle and secret logic, and Loot Integrations (optional) to push modded gear into modded structure chests.

Magic structures get the same five things, with one addition: their secrets are the only source of **Eldritch Manuscripts**, so the school-gated content sits behind the same discovery loop as everything else.

Budget & cut order for this layer

Core (17): YUNG's Better Dungeons · Strongholds · Mineshafts · Ocean Monuments · Nether Fortresses · Desert Temples · Jungle Temples · Witch Huts · End Island (+ YUNG's API) · Repurposed Structures · Hopo Better Ruined Portals · Hopo Better Underwater Ruins · The Graveyard · Structory · Dimensional Dungeons · Loot Integrations. Optional (2): When Dungeons Arise (now largely redundant — **cut it**; Structory +

YUNG's + Graveyard + Hopo cover the ground), YUNG's Bridges / Extras.

Cut order if perf or worldgen time suffers: When Dungeons Arise → Structory → Hopo Better Underwater Ruins → Repurposed Structures → YUNG's End Island → YUNG's Witch Huts. YUNG's Better Dungeons / Strongholds / Mineshafts are cut last.

Conflict notes: YUNG's Better Mineshafts and Hopo Better Mineshaft both *replace* mineshafts — **use YUNG's only**. Repurposed Structures *adds* variants rather than replacing, so it coexists with YUNG's, but its per-structure spawn rates and dimension/biome blacklists must be configured so the world isn't flooded. Structure mods raise chest and spawner counts, which is a CPU/entity cost, not a GPU one.

6. Mod stack

All verified for **1.21.1 NeoForge** (§16). Versions marked (**pin at build**) are locked at assembly. **Mod budget: ≤130 content (incl. QoL/feedback) + ≤18 performance + ≤20 libraries** (*raised from 120 when the T1 feedback batch and the magic addon were approved — §13's cut order is the pressure valve*).

The mod stack is generated, not hand-maintained. `tools/modlist.toml` is the source of truth; `python tools/build.py sync` resolves every version and sha512 from Modrinth and writes the packwiz project (`pack/index.toml` + `pack/mods/*.pw.toml`). **131 of 132 mods resolve automatically** (118 Modrinth + 12 CurseForge + 1 GitHub); the only one left is **GhostSwap**, our own custom mod, due in M1. Six further mods were dropped after research found no 1.21.1 NeoForge build — see `[dropped]` in `tools/modlist.toml`. See PIPELINE.md §1.5 and README.md.

6.1 Core libraries

NeoForge 21.1.x · **Curios API 9.4.2+** (Cataclysm, Enigmatic Legacy) ·

Lionfish API 3.0 (Cataclysm) · Citadel (Alex's Mobs ports) · Patchouli · Architecture API · Kotlin for NeoForge · Terrablender/Biolith · Player Animator · YUNG's API (if YUNG's kept)

6.2 Combat & movement

Mod	Role
<i>(vanilla combat — deliberately no overhaul mod)</i>	
Backported Spears 1.8.0	Spear: jab + speed-scaled charge + Lunge dash + Mob Kabob advancement + <code>piercing_weapon</code> component
Combat Roll 2.0.6+1.21.1	Dodge roll, i-frames, attributes/enchants
ParCool 3.4.3.x	Parkour movement + stamina
ParCool+ Compatibility++	Animation compatibility layer
Custom "GhostSwap" mod <i>(we build)</i>	Attribute swapping automation, toggle key

6.3 Gear & stats

Mod	Role
Attribute Modify (NeoForge 1.21.1)	Datapack-driven attribute tuning for vanilla + Curios items
Reforged (Tiered fork)	Crafted gear gets tiers + modifiers; stat rerolling at anvil with hammers
Simply Swords	Spear, glaive, chakram, katana, greathammer, rapier, twinblade + uniques

Spartan Weaponry (Unofficial)	Polearms, throwing knives, crossbows, quivers, weapon oils (<i>disable its spears — we have real ones</i>)
Cataclysm gear	Witherite weapons + Mechanical Fusion Anvil upgrade path
Alex's Caves gear	Darkness Armor, Dreadbow, Raygun, Seeking Arrow

6.4 Difficulty & soulslike

Silent's Power Scale (*replaces Scaling Health — no 1.21.1 build for the latter; same author, generalises it to any attribute on players and mobs*) · **Mob Champions** · GraveStone Mod · custom Ember Flask (KubeJS)

Mob Champions — the Minecraft Dungeons *enchanted mob* idea. Any whitelisted hostile spawns in four ranks (Uncommon → Rare → Epic → Legendary) with configurable stats and effects, dropping rare and normally-unobtainable loot. Whitelisted to hostile overworld/Nether mobs; stats capped so **distance scaling still dominates** — a champion near spawn must never out-threat the Reaches.

The Ashenfall Threat — the *Apocalypse / Apocalypse Plus* slider. Seven tiers (I–VII), selectable per region, each raised by **clearing that region's bosses at the current tier**. Higher threat = more champions, harder scaling, rarer loot. This is the answer to "how does an infinite world stay dangerous", and the long-term reason to return to a region you've already cleared.

6.5 RPG & grinding

Mod	Role
Project MMO	Skills with real attribute effects
Project MMO: Classes	Class system
Project MMO: Skill Books	Skill books as boss/dungeon loot

Project MMO: XP Bottles	Bankable XP as loot/drops
Origins (NeoForge) + Origins: Classes	Origin/class fantasy layer
Mob Grinding Utils	Fans, absorption hoppers, masher — real farms for materials

Grinding philosophy: grinding is always meaningful, never a wall. Skill levels give real stats; skill books drop from bosses; XP bottles bank progress; farms are buildable; repeatable quests turn grinding into rewarded checklists.

Classes (5; respec is expensive so the choice matters): Vanguard (poise/shield/aggro) · Reaver (melee are buildable; repeatable quests turn grinding into rewarded checklists.

Classes (5; respec is expensive so the choice matters): Vanguard (poise/shield/aggro) · Reaver (melee burst, dual-wield) · Ranger (range/mobility/crits) · Ember-Warden (flask efficiency, healing, anti-undead) ·

Hexer (spellcasting, mana economy, spell-imbued weapons — starts at -40 with the Witchbane)

6.6 Bosses & dungeons

L_Ender's Cataclysm (primary spine: Ignis, Netherite Monstrosity, Ender Golem, Leviathan, Ancient Remnant, The Harbinger, Scylla, Maledictus — two-phase fights, hand-built arenas, Witherite →

Mechanical Fusion Anvil) · **BOMD** (Night Lich, Obsidilith, Nether Gauntlet) · **The Twilight Forest** (Naga, Lich, Hydra, Ur-Ghast, Final Boss) · **The Aether** (Slider, Valkyrie Queen, Sun Spirit) · **Deeper and Darker** (Otherside) · **Alex's Caves (Unofficial Port)** · **Iron's Spells 'n Spellbooks** (core — see the magic layer below) · **Monsters & Spellbooks** (the Necro school — approved, highest-risk item, cut first)

The magic layer — Iron's Spells 'n Spellbooks

Magic is a **full parallel track**, not a bolt-on. It has its own progression chain, its own gear, its own bosses, and its own faction politics — and

the story already assumed it existed.

What it adds: nine spell schools (Fire, Ice, Lightning, Holy, Ender, Blood, Evocation, Nature, **Eldritch**) · a mana economy (max_mana, mana_regen, spell_power, cast_time_reduction, cooldown_reduction, per-school spell power and magic resistance) · five craftable spell book tiers (iron 6 slots → diamond 10 → netherite 12, expandable to 12) plus unique books including the **Necronomicon** (dropped by the Dead King, +2 levels to Raise Dead, 10 slots, +200 mana) · ten school armour sets that add max mana and boost one school's spell power · wizard NPCs (Pyromancer, Cryomancer, Priest, Apothecarist, Necromancer, Archevoker) that trade · and three bosses with their own structures: **the Dead King** (Catacombs), **the Keeper** (Lost Citadel), **the Fire Boss** (Echo of Tyros).

Its three workstations join the crafting chain: **Scroll Forge** (craft scrolls from paper + a school focus + ink of a chosen rarity) → **Inscription Table** (inscribe scrolls into a spell book, and extract them again without losing levels) → **Arcane Anvil** (upgrade scrolls, upgrade armour, and imbue a weapon with a spell).

Why it belongs in this pack:

- **The Arcane Anvil's weapon imbuing** is the single best fit in the whole mod list: every signature boss weapon gets an imbued-spell slot, so a Knightmetal katana can carry Fang Ward and a Witherite greathammer can carry Blaze Storm. Build diversity without touching vanilla swing mechanics.
- **The Eldritch school is lore-gated:** casting it requires an **Eldritch Manuscript**, consumed on use, found only in late-game structure chests or dropped by Eldritch-themed enemies. That is exactly our design language — a whole school of magic behind a discoverable gate.
- **The Witchbane Crusaders' hatred of magic is now playable**, not flavour text: a Hexer starts at -40 standing with them and has to decide whether that matters.
- **Wizard NPCs trade** — the one "villager" economy we can allow, because they sell *materials* (arcane essence, ink, focuses, runestones) rather than vanilla power gear. **Their scroll trades get stripped** to keep rule 9 intact.

- **Casting has real costs:** mana, cast times and cooldowns mean a Hexer must manage positioning and stamina-adjacent resources — very soulslike.

Progression hook: arcane essence from spellcasters → Scroll Forge → school armour → raid magical structures for **runestones** → Arcane Anvil → netherite spell books → Eldritch → the three magic bosses.

The tenth school — Necro (approved). Monsters & Spellbooks

(1.21.1 NeoForge, 0.0.16.3) is **in**, not optional: it adds the **Necro school** (debuffs, DoT, curses), 90+ spells, magic-armed enemies and Arch/Hybrid armour tiers. Necro is *not* one of the Empire's nine — it is older, and it is what the Seal-bearers imprisoned in the first place, which is why it pairs perfectly with The Graveyard's lich and why the Witchbane's fear isn't entirely paranoid. **It is still the highest-risk item in the stack** (actively developed, likely reworked), so it gets its own compatibility pass in M3c and is the first thing to cut if it destabilises — but it ships.

6.7 Nations & factions

Millénaire (1.21.1 NeoForge) — seven living cultures (Norman, Indian, Japanese, Mayan, Byzantine, Inuit, Seljuk Turk) with architecture, trades, currency, villager lives, village growth, **reputation & diplomacy**, culture quests, village chronicles and raids. **Ancient Warfare 2 (dropped)** — no 1.21.1 build. The nation set-pieces it was going to supply are carried by Towns and Towers, Structory and Repurposed Structures; the nation *identities* are carried by Millénaire's seven cultures plus the KubeJS standing system. Plus a custom **standing system** in KubeJS + FTB Quests.

Risk flag: Millénaire's 1.21.1 port is community/alpha and 9.0.2 has only ~1.2k downloads — treat it as unproven and keep **D7** open, with the fallback (vanilla villages + Towns and Towers + KubeJS-only factions) fully specified. **AW2 is out entirely** — no 1.21.1 build exists.

Scaling Health is also out (no 1.21.1 build). Replaced by **Silent's Power Scale**, by the same author, which generalises the idea: a scaling-attribute system that can modify *any* attribute on players *and* mobs. This is what carries the distance/depth difficulty curve, so the

loss is fully covered.

6.8 Loot, items, collectibles

Lootr · Artifacts (~20 curios) · Enigmatic Legacy+ (*unofficial port*) · boss music discs (Cataclysm) · **Loot Beams: Refork** (rarity beams on world/chest drops) · all of §6.3

Rarity is visible at a glance — Common → Uncommon → Rare → Epic → Legendary, each with a beam and colour, on ground drops and in chests. It makes Reforged's modifier tiers and Mob Champions' loot legible without opening a menu, which is the single biggest readability win in the pack.

6.9 Questing & narrative

FTB Quests (NeoForge) · FTB Teams / FTB Chunks · Patchouli (The Pilgrim's Journal) · KubeJS (custom items, standing, dialogue, puzzle logic)

6.10 QoL

JEI + Jade + AppleSkin · Mouse Tweaks · Controlling (*essential with this many keybinds*) · Carry On · Xaero's Minimap + World Map · Fast Travel Waypoints · Waystones · Explorer's Compass / Nature's Compass · Inventory Profiles (*optional*) · Shutup Experimental Settings! · Skin Layers 3D · Ambient Sounds 6 · Sound Physics Remastered · Presence Footsteps · Dynamic FPS · **YUNG's Traveler's Titles**

6.10a Feedback & juice (approved from RECOMMENDATIONS T1)

Mod	What it does for this pack
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Mini-boss Boss Bars	Boss bars on the mini-bosses the big boss-bar mods ignore — Hollow Challenge champions, nation champions, Mob Champions at Epic rank. Explicit Alex's Caves integration
Configurable Boss Bars	Client-side styling of every boss bar, so a 21-boss gauntlet reads consistently instead of every mod shipping its own
Floating Damage Indicators	Outgoing + incoming damage, colour-coded by damage type — the readability layer for stamina-governed combat. <i>(Mutually exclusive with Damage Numbers Mod — this one wins.)</i>
Boss Music Mod	Per-boss themes for all 21 bosses. <i>(Mutually exclusive with Fight Music and Dash's Custom Background Music Changer — pick this one and supply the audio; Dash's stays the fallback if its MP3 workflow proves easier.)</i>
Loot Beams: Refork	See §6.8 — rarity beams on drops and chests

Why these were added: a 21-boss gauntlet with 12 combat trials needs its feedback legible. Every one of these is client-side or config-only, so none of them touches the balance or the worldgen.

6.11 World fullness layer — making the world feel lived-in

These mods exist for one reason: so the world feels *whole* rather than generated. Every one of them is also hooked into an existing system — none of them is decoration-only.

Mod	What it adds	How it hooks into the pack
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FallingTree	Fell an entire tree by cutting one log. Modes: instantaneous / shift-down (clears the trunk from the ground) / falling-block animation. Whole-tree detection scans all directions. Sneak + left-click disables it, so your own builds are safe. Works with any modded logs via <code>minecraft:logs</code> tags	Woodcutting skill gate: the number of logs you can fell in one cut scales with your Project MMO Woodcutting level — so tree felling <i>is</i> the early grinding loop, and Terralith's huge forests become a real resource economy
Star Worm Equestrian (SWEM)	The "better horses" mod: custom horse model + animations, separate gaits (walk/trot/canter/gallop), training and bonding, coat colours and genetics, full English/Western/Adventure tack (bridles, saddles, girths, blankets, leg wraps, breastcollars), saddle/bridle/ armour racks, horse illnesses	Mounted combat is a real build. The backported spear's charge damage scales with <i>relative speed</i> — SWEM's gaits and genetics make "breed the fastest horse in the Reaches" an actual goal. Nation horse lines: Norska warhorses, Seljuk desert steeds, Norman destriers as nation-gear rewards

Sophisticated Backpacks (+ Sophisticated Core)	Placeable, colourable backpacks with inventory upgrades and a deep upgrade system: tool swapper, feeding, filters, fluid tank, mob catcher, void upgrades, JEI/Curios integration	Progression-gated carry weight. Backpack crafted at Tier 1; upgrades drop from bosses (Ignis → tank upgrade; Night Lich → filter upgrade) and from standing rewards. The mob catcher powers a Bestiary collection quest — "items to collect" now includes creatures
Guard Villagers	Villages spawn guards (configurable count); guards fight raids; clerics heal guards, villagers and players with Hero of the Village (3×/day); golems repaired by smiths once a day	Raids are faction events. Defending a Norman Remnant village from a pillager raid = +standing with the Remnant. Letting it burn = –standing. This turns vanilla's most annoying event into the story's main social lever

Farmer's Delight	New crops (tomato, cabbage, onion, rice), rich soil, cutting board + knife, stove cooking, high-saturation prepared meals, rustic decoration	Food is a system, not a chore. Cooked meals give real saturation and short buffs; the Ember-Warden class specialises in them. Taverns (Dungeons and Taverns) become places you actually want to stop
Supplementaries <i>(optional)</i>	Large vanilla+ block/item set: ash, briquettes, cages, cannons, lanterns, sign posts, urns, safes, ticking clocks, rope, and more	Gives you the blocks to build a base that belongs in this world; cannons give the Norska siege flavour. First mod cut if FPS suffers
Comforts <i>(optional)</i>	Sleeping bags and hammocks	More rest points = more bonfires = more flask refills. Pairs with the Waystone network
Tool Belt <i>(optional)</i>	A belt slot holding up to 3 tools	QoL; synergy with the backpack's tool swapper

Dynamic Waters	Rivers are <i>carved into terrain during world generation</i> — water flows in set directions, mountain streams run down into river valleys, branching waterways and coastal deltas form. Boats and items are carried by the current. Verified compatible with Terralith, Regions Unexplored, BOP, Nature's Spirit	The world's rivers become real geography. "The tower three days north of the frozen river" is now an actual landmark you can follow. River mouths and deltas become the natural homes of coastal nations and harbours. Note: requires a brand-new world — the water stack must be locked before world generation.
Flowing Fluids	Realistic fluid physics: water flows downhill and spreads, a fluid pressure system (fluid with fluid above it spreads faster and further, propagating downward and sideways), mostly-finite water, and per-entity-type control over whether height differences push boats/items	Water is terrain, not decoration. Dams, canals, flooded ruins and draining a vault become real engineering. Also the hazard layer: falling water, currents and pressure in the deep caves
Boatload (optional)	Alternative sailing — larger/cargo vessels beyond the vanilla boat	Seafaring for the coastal nations; pairs with the backpack's storage for ocean expeditions

Sable + Sablewaves <i>(optional, perf-gated)</i>	A physics engine plus an ocean layer: visible moving sea surface, ocean currents that carry your ship, moon-driven tides, weather-driven swells that grow in thunderstorms, and wind that heels floating craft	The ocean becomes a place. This is the "oceans have currents" ask in full. Sablewaves also understands vessel drift speed and integrates with Sable/Physics-mod vessels. Cost: a physics engine is CPU-heavy — see the budget rule
Currents of Trade <i>(optional)</i>	Vanilla-friendly maritime trade network: certified harbours at coastal villages, a 3D sailing sloop, autonomous cargo fleets between harbours	Story-perfect for the Guild of Merchants: their trade routes become literal ocean currents, and connecting coastal villages into a network is a standing-reward activity
Streams Reflowing <i>(alternative to Dynamic Waters)</i>	Streams-style rivers that carry boats and items, with a <code>vesselDriftSpeed</code> setting and Sable vessel support; overworld-only	Use this instead of Dynamic Waters if the two conflict — pick one river mod, not both

Budget rule: the fullness layer is capped at **7 core mods + 3 optional**, plus a separate **water & sailing layer of 3 mods** (Dynamic Waters + Flowing Fluids + Boatload, or Streams Reflowing in place of Dynamic Waters). If FPS or tick-time drops below target, the cut order is: Currents of Trade, then Sable/Sablewaves, then Streams Reflowing, then Supplementaries, Comforts, Tool Belt, Farmer's Delight, Boatload, Flowing Fluids, and SWEM last. SWEM and Dynamic Waters are cut

last because mounted combat and real rivers are design pillars — though if SWEM must go, horses stay vanilla and the spear's charge still scales with elytra/minecart/falling speed.

The water & sailing layer — three things it buys us

1. **Rivers are landmarks and roads.** Dynamic Waters carves real river networks at worldgen, so directions given in rumours ("three days north of the frozen river") point at something you can actually follow. River valleys become the safest long-distance travel routes — and the most dangerous ambush spots.
2. **Boats are a combat platform.** The spear's charge attack scales with *relative speed*. A boat running a downhill current with a spear braced is a jousting weapon — the single best argument for the water layer existing in a combat pack.
3. **The ocean is where the story's money lives.** With Sablewaves' currents and tides, and Currents of Trade's harbour network, the Guild of Merchants' trade routes become a literal mechanic you can sail, raid, or join.

Perf note: fluid ticks are a *CPU* cost, not a GPU cost, so your 3500-series card is fine — but Flowing Fluids ships performance/visual/gameplay settings and Dynamic Waters precomputes river paths at worldgen specifically to avoid runtime freezes. Both must be configured rather than left on defaults, and large water volumes (oceans, dams) are the thing to watch in testing. Dynamic Waters also requires a **new world**, so this stack is locked before M3 worldgen.

6.12 Performance

Core: Embeddium + Embeddium Extra/++ · ModernFix (dynamic resources + faster item rendering **on**) · FerriteCore · ImmediatelyFast (experimental_screen_batching **on**) · Entity Culling · Clumps · Alternate Current · Enhanced Block Entities (NeoForged) · FastSuite · MoreCulling · Structure Layout Optimizer · Dynamic FPS **NVIDIA (your card): Acedium Sodiumized** — NeoForge 1.21.1 port of Nvidium; requires GTX 1600+/RTX 20xx+. Big terrain-rendering win at high render distance. Disables itself under shaders and falls back to stock

Embeddium if extensions are missing. **World: Chunky** for starter-region pregen and on-demand pregen.

Magic perf note: spell effects are particle-heavy. Cap the pack's particle settings, and if the Hexer track drags FPS in a boss arena, the first lever is Iron's Spells' own particle settings — never the render-distance budget.

7. Story — The World of Vantyra

7.1 Premise

Twelve centuries ago the **Vantyrlic Empire** ruled the world, held together by **Nine Ember Seals** — nine champions, each bearing an Ember, each holding back the **Ashenfall**: a blight that unmakes the world, mobs that don't burn in daylight, a sky that slowly dims.

The Empire's scholars — the order that became the **Byzantine Choir** — learned to *forge* Embers in the **Ember Forges**, and did so without the Emperor's leave. The **Tenth Ember** was flawed.

The Tenth Ember shattered the Nine. The Empire fell in a single night. Its capital became the **Cathedral of Ash**, and the Ashenfall rolled outward — and never stopped. It simply thins with distance.

The survivors rebuilt as the **Nine Nations**, each inheriting a province, each keeping a **Seal Shard**, and each telling a *different* story about who destroyed the world.

Now the Seals are failing again. The sun is dimmer than it was. There are things in the dark that used to fear it.

7.2 The player

A pilgrim, found drowned on a cold shore with no memory and a **cold**

Ember in the chest. That Ember is the Tenth — reforged, dormant, and the only thing keeping you alive. Your **Ember Flask** is its remnant. Every boss you defeat feeds it.

You are not the chosen one. You are the *cause*, walking around trying to fix what you broke.

7.3 The Nine Nations

#	Nation	Anchor (mod)	Territory	Attitude	What they offer	The cham
1	The Norman Remnant	Millénaire: Normans	Heartlands	Neutral, trade-hungry	Waystone network, repair, quest hub	Lord of the Remnant
2	The Seljuk Expanse	Millénaire: Seljuk Turks	Deserts/badlands	Wary; hostile if you loot ruins	Horse gear, ranged weapons	Desert Warlord
3	The Byzantine Choir	Millénaire: Byzantines	Old imperial roads	Coldly curious — they know what you are	Lore , seal knowledge, the spell tier	The Archivist
4	The Mayan Deep	Millénaire: Mayans	Jungles & mesas	Ritualistic; demands tribute	Blood/alchemy gear, potions	The Sinner Eater
5	The Yamato Shogunate	Millénaire: Japanese	Bamboo highlands	Respects strength only	Katana/twinblade tier	The Daimyo
6	The Inuit North	Millénaire: Inuit	Frozen wastes	Pragmatic survivors	Cold-weather gear, Cold Tower intel	Frost Warlord

7	The Spice Kingdoms	Millénaire: Indians	River valleys	Mercantile, gossip-rich	Alchemy, flask upgrades	The Mahar
8	The Norska Clans	KubeJS-only (<i>no AW2 build</i>) — pillager-outpost retexture + a Towns and Towers coastal village set	Northern coasts / Nether-adjacent	Raiding; ally by strength	Siege weapons, berserker gear	The J
9	The Witchbane Crusaders	KubeJS-only (<i>no AW2 build</i>) — a Structory fortified-march structure plus a KubeJS faction	Fortified marches	Hostile to magic users; they know the truth	Anti-magic gear, the real history	The Grand Inquis

Unaffiliated powers: The Guild of Merchants (KubeJS-only, anchored on Millénaire trade caravans — neutral, everything has a price, and they've been *hoarding the Tenth Shard* to sell it) · Brigands / Pirates / Amazons / Barbarians (vanilla illager variants + Structory camps) · **The Hollow Court** (Twilight Forest — the Empire's lost province, frozen mid-collapse) · The Aether's Valkyries & Sun Spirit (the Ember Forges were built using Aetherite; they want their property back).

7.4 Faction standing

- **Range:** −100 (hunted on sight) → +100 (champion of the nation).

- **Driven by:** quest choices, tribute, raiding their structures vs defending them, sparing vs killing their champions, class (a Hexer starts at -40 with the Witchbane).
- **Gates:** $\geq +30$ unlocks that nation's gear line + champion quest; ≤ -50 makes their patrols hostile and closes traders; $+80$ unlocks a nation-exclusive ultimate weapon.
- **Cross-faction:** helping one nation's enemy costs standing with them. You cannot be champion of all nine — *which is what makes the ending choice real.*

7.5 Three-act plot

ACT I — "The Cold Shore" (Ch 0–3) Survival, first contact with the nations, first bosses. Ends with the reveal that the **Night Lich is the first Seal-bearer**, preserved as a lich by the fall; killing it releases its Seal Shard into your Ember, and every nation in the world feels it. The Byzantine Choir sends an Archivist. The Guild of Merchants makes an offer.

ACT II — "The Nine" (Ch 4–6) Travel the nations, earn standing, hear each nation's contradictory account of the fall. Each champion holds a Seal Shard. The Hollow Court is the lost province. **Twist:** the Witchbane Crusaders are right — the Ember Forges caused the fall; and the Guild of Merchants has been sitting on the Tenth Shard for a century, waiting for the price to be right. Expose them, buy from them, or rob them.

ACT III — "Ashenfall" (Ch 7–8) The End is the Empire's capital. The **Harbinger** is the Emperor's herald, still waiting for a liege. The Gauntlet is the four champions who held the line. **Final revelation:** Maledictus is the last Emperor's champion, still holding the Nine Seals together by sheer will — and he is *tired*.

Three endings:

1. **The Empire Rekindled** — take the Seals, rebuild Vantyra, become its emperor. (Requires high standing with the Choir.)
2. **The Pilgrim's Road** — scatter the Seals; the nations rise or fall on their own. (Requires high standing with ≥ 3 non-Choir nations.)
3. **The Tenth Ember** (*hidden*) — forge a Tenth Ember properly: all

24 journal pages, one nation at +100, and the Hollow Challenges cleared. You stop the Ashenfall and become the new Seal-bearer — bound to the world forever. Hardest to earn, and the only one where you live.

- 4. **The Ashen Throne** (*secret — the rune route*) — assemble all nine Ember Runes, open the Hidden Cathedral, and take the Ashen tier's core material for yourself instead of using it to forge a Seal. It is not on the critical path, it requires no standing with anyone, and it is the ending for players who explored instead of followed the road.

8. Puzzles & Ancient Lore (USW's core)

This is the section that makes the world feel *designed* rather than generated.

8.1 The Annals — lore as a lock

The Empire's Annals: 12 ancient texts scattered across the world (one per act/region). They must be found **in order**; each contains a fragment of the fall *and* a clue to the next. Collecting all 12 reveals the true location of the **Cathedral of Ash** — which is otherwise unfindable. *This is the ending gate.*

8.2 Puzzle types (built, not modded)

Puzzles are implemented as **custom datapack structures + KubeJS logic**, so they're guaranteed to work and to fit the lore:

Puzzle	Where	Mechanic
Riddle Vaults	Empire ruins, nation capitals	Patchouli lore gives a riddle; a KubeJS "code lock" block accepts the answer as an item or a written-book submission

Elemental Gates	Ember Forges, Aether dungeons	Sealed doors that only open with the right protection: fire-proof gear, water breathing, feather falling, a specific potion
Pedestal Ordering	Seal Vaults	Place the correct Seal Shards/ artefacts on pedestals in the right order — order learned from the Annals
The Labyrinth	Hollow Court	Twilight's native maze + a custom "thread" item that marks the true path
Parkour Trials	Every landmark	USW-style obstacle courses using ParCool — vaults, wall-runs, ledge grabs. Fail = fall into a hazard, not a reset
Combat Trials	12 Hollow Challenge arenas	Waves with restrictions (no armour, one flask, timer)
Lever/Song Puzzles	Cathedral of Ash	Vanilla redstone + custom controllers, solution hidden in music-disc lore entries

Mods that add native puzzle content and are used where they exist: **The Twilight Forest** (push-block towers, light puzzles), **The Aether** (locked dungeon doors, switch sequences), **Cataclysm** (arena mechanics that must be read), **Dungeons and Taverns** (hidden rooms).

8.3 The Ember Runes — the secret mission

Minecraft Dungeons hides a **secret mission behind hidden runes**, and this pack already has the perfect equivalent. **Nine Ember Runes** are hidden in nine secret areas across the world — one in each of the nine nations' capitals, each behind a riddle, a parkour route or a mini-boss that only exists if you go looking. Each rune is a single item with no recipe and no JEI entry; you will not know what it does until you have two.

Find all nine and the Hidden Cathedral opens: a secret, fully optional endgame dungeon placed somewhere the map does not show, holding the only source of the **Ashen** crafting tier's core material and one of the three endings. It is not on the critical path, it is not soft-locked, and no compass can find it — only the runes themselves, which tell you where to go once assembled.

This is the pack's strongest "the world rewards exploring properly" mechanic, and it costs nothing but custom items, a structure and KubeJS logic.

8.4 The Pilgrim's Journal (24 pages)

Hidden across all regions — dungeon chests, boss drops, village libraries, buried in landmark structures. Each page is a fragment of the fall, written by someone who was there. Several pages *are* treasure maps.

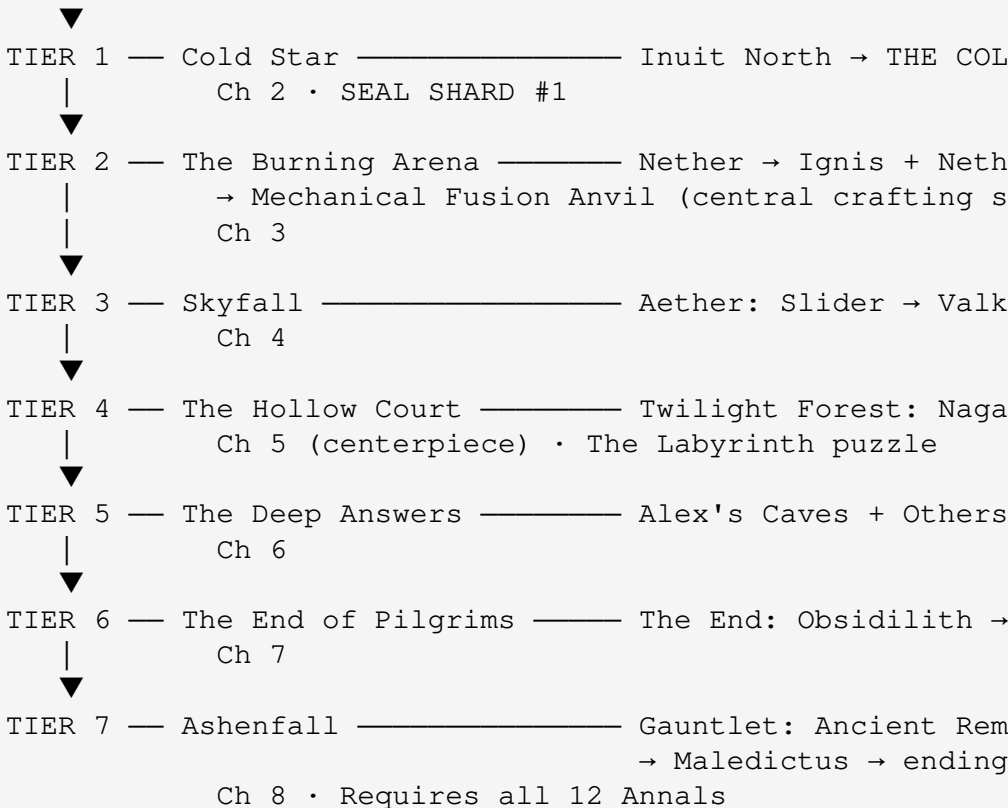
8.5 Environmental storytelling

Every structure gets a "who built this and why." The Empire's provinces are named onto Towns and Towers and Structory output in KubeJS — a `structures.js` province register that maps a generated structure to a fallen province, its nation and its loot theme. Millénaire village chronicles record births, deaths, raids and building works — living history you can read. Boss music discs carry named in-world lore entries.

9. Progression spine

Seven tiers; each tier's entry point is a **discovered landmark**, not a tier number.

TIER 0	—	Pilgrim's Road	—————	Heartland (Norman Rem
		Ch 0-1	• Landmarks:	Drowned Chapel



MAGIC TRACK — (runs in parallel, any point after Tier 2)
Arcane Essence from spellcasters
└─ Scroll Forge → Inscription Table → school armour
└─ raid magical structures for RUNESTONES
└─ Arcane Anvil (scroll upgrades · armour u
└─ netherite spell book → ELDRITCH (r
└─ Dead King → Keeper → Fire Boss

10. Boss gauntlet

#	Boss	Mod	Location	Gimmick	Reward / gate
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1	Night Lich	BOMD	Cold Tower	Projectile volleys + teleports; cover pillars	Seal Shard #1 , Tier-1 boss weapon
2	Nether Gauntlet	BOMD	Nether structure	Melee rushdown, punishes panic-rolling	Tier-2 mat
3	Ignis	Cataclysm	Burning Arena	Fire phases; shield-break with axe crits	Mechanical Fusion Anvil + Witherite
4	Netherite Monstrosity	Cataclysm	Nether arena	Charges, lava phase, ground slam	Witherite tier
5	Slider	Aether	Bronze dungeon	Knockback focus, arena hazard	Aether gear
6	Valkyrie Queen	Aether	Silver dungeon	Air phase, lance charges	Gravitite access
7	Sun Spirit	Aether	Sun altar	Eternal-day mechanic, beams	Aether endgame gear
8	Naga	Twilight	Forest clearing	Segment combat, arena circling	Hollow Court progression

9	Lich	Twilight	Tower	Multi-phase, adds, shield juggling	Ironwood/ Steeleaf tier
10	Hydra	Twilight	Swamp/ forest	Multi-head, head regrowth	Knightmetal tier
11	Ur-Ghast	Twilight	Dark tower	Vertical, tear-based	Hollow Court endgame gear
12	Obsidilith	BOMD	End islands	Laser patterns, mobility check	End tier mat
13	Ender Golem	Cataclysm	End arena	Charge + shockwave	End gear tier
14	The Harbinger	Cataclysm	End arena	Flying, multi-phase, high DPS check	Gauntlet Key
15	Ancient Remnant	Cataclysm	Ruined city	Hardest melee check	Endgame weapon
16	The Leviathan	Cataclysm	Ocean	Water combat, ranged	Endgame weapon
17	Scylla	Cataclysm	Acropolis (sky)	Storm-phase arena	Endgame weapon
18	Maledictus	Cataclysm	Cathedral of Ash	Final encounter	Ending choice

19	Dead King	Iron's Spells	The Catacombs	Summon phases; the arena fills with the risen	Necronomicon (+2 Raise Dead, 10 slots, +200 mana)
20	The Keeper	Iron's Spells	The Lost Citadel	Arena control; punishes standing still	Imbued-weapon materials, cooldown gear
21	The Fire Boss	Iron's Spells	Echo of Tyros	Beam patterns, add waves	Fire-school legendary scroll

Plus **nine nation champions** (standing-gated) and **12 Hollow Challenge combat trials**.

Plus the magic gauntlet: the **Dead King** (Catacombs) → **the Keeper** (Lost Citadel) → **the Fire Boss** (Echo of Tyros). These three are not chapter bosses — they are the Hexer track's gauntlet, run in parallel, and the Dead King drops the **Necronomicon**.

11. Weapons, gear & crafting

Tier ladder: Iron/Diamond → Netherite → **Witherite** (Cataclysm) → Aether (Zanite/Gravitite/Ambrosium) → **Arcane** (spell books, school armour, runestones — Iron's Spells) → Hollow Court (Ironwood/Steeleaf/Knightmetal) → Alex's Caves → Cataclysm endgame → **Ashen** (custom).

Crafting stations (milestones): vanilla smithing → **Mechanical Fusion Anvil** (Cataclysm) → Aether Altar/Freezer → **Scroll Forge / Inscription Table / Arcane Anvil** (Iron's Spells — the magic chain) → Hollow Court infusion → custom **Ashen Anvil** (KubeJS).

Weapon families: vanilla (sword/axe/mace/trident/bow/crossbow) ·

spear (Backported Spears — the new movement weapon) · Simply Swords (glaive, chakram, katana, greathammer, rapier, twinblade) · Spartan Weaponry (polearms, throwing knives, crossbows) · Alex's Caves (Dreadbow, Raygun).

Signature weapons: one per boss (~12), each with a gimmick, each upgradeable through the tier ladder using the *next* boss's material. Every signature weapon has an **imbued-spell slot** (Arcane Anvil) — this is where melee builds and magic builds meet. Plus one **nation-exclusive ultimate** per nation at +100 standing.

Gear depth: Attribute Modify tunes every stat to vanilla-consistent numbers; Reforged adds craftable tiers/modifiers with anvil rerolling.

11.1 Where gear comes from (villagers are stripped)

This deserves a direct answer, because "no villager trades" sounds like "no gear". It is the opposite: **gear comes from your own hands, from danger, and from the world — never from a shop.**

Source	What you get there	Notes
Your own crafting (the main path)	Everything, eventually. Material tier ladder + crafting stations	Smithing skill level gates each recipe. Vanilla smithing, then Mechanical Fusion Anvil (from Ignis), Aether Altar/Freezer, Hollow Court infusion, Ashen Anvil . You gather materials by mining, farming, felling, fishing and killing
Dungeons and structures	Artifacts, enchanted books, skill books, cartographer's maps, materials	Lootr chests in When Dungeons Arise, Dungeons and Taverns, YUNG's, nation capitals and ruins. This is the loot faucet that replaces villager trading

The magic track	Spell books, school armour, scrolls, runestones, arcane essence	Made and found, never bought: Scroll Forge → Inscription Table → Arcane Anvil. Runestones and the Eldritch Manuscript come only from magical structures and Eldritch-themed enemies. Wizard NPCs sell <i>materials only</i> — their scroll trades are stripped, because you cannot buy power
Bosses	Signature weapons + the material that forges the next tier	21 bosses, each with a unique weapon and an upgrade path into the next tier
Mobs	The armour and weapons they are wearing, plus their drops	Distance/level scaling keeps early drops relevant; Silent's Power Scale raises mob attributes with distance, and pillager outposts supply real equipment
Enchanting	Vanilla enchanting table + bookshelves, fully intact	Enchanted books come from loot chests, fishing treasure, and piglin bartering — bartering is <i>kept</i> , because the Nether is dangerous, so it is an activity rather than a shop
Faction standing	Each nation's gear line, and a nation-exclusive ultimate at +100	Earned by quests, tribute, and defending their villages
Millenaire culture trades	Culture-specific goods: food, tools, culture gear, quests — bought with that culture's own currency	Villages still trade. What is stripped is vanilla power trades: no diamond/netherite gear, no enchanted books, no potions

The Guild of Merchants	Anything, for a price	Story-flavoured, expensive, morally suspect — and they are hoarding the Tenth Shard
Reforged hammers (the one villager item that matters)	Three hammer tiers sold by villagers, used to reroll gear stats at an anvil	A reroll currency, not gear — so villagers affect <i>quality</i> , never <i>availability</i>
Repair	Smiths repair; iron golems repaired by smith villagers once a day	Guard Villagers' clerics will heal you if you hold Hero of the Village

The short version: **villagers stopped being a gear shop so that gear has to mean something**. Every piece you own was mined, farmed, felled, killed for, found, or earned from a faction — and the good stuff can only be *crafted* once you have beaten the thing that drops its material. **Recipe gating (KubeJS)**: modded recipes above Tier 1 require Project MMO skill levels; boss-material gear requires the matching FTB Quests flag; vanilla recipes rebalanced expert-style (no villager power, netherite needs a wither skull).

12. Performance & GPU plan (your 3500-series card)

Budgets

Metric	Target
FPS	60+ at render distance 10–12, no shaders
RAM	6 GB (8 GB max)
Mods	≤ 120 content + ≤ 18 utility

Entities near player	Capped low — Alex's Mobs / Millénaire villagers are the biggest tick cost
World	Infinite; starter region pre-generated with Chunky (~10k × 10k), on-demand pregen after

The wins, in order

1. **Embeddium + Acedium Sodiumized**. Embeddium replaces the renderer; Acedium (Nvidium port, GTX 1600+/RTX 20xx+) pushes terrain rendering onto your NVIDIA GPU. This is the single biggest FPS lever on your card.
2. **ModernFix + FerriteCore + ImmediatelyFast** with the specific config flags flipped.
3. **Starter-region pregen + on-demand pregen**. You never watch chunks generate during the story; you pregen before long trips.
4. **Entity caps + structure density tuning**. When Dungeons Arise and Structory are the two heaviest structure contributors; both get tuned down.
5. **No shaders, ever**. Acedium and Iris/Oculus disable each other. Mood comes from Ambient Sounds, Sound Physics Remastered, Presence Footsteps instead.

Profiles

- **performance** — Acedium on, render distance 10, entity caps tight.
- **compat** — Acedium off (stock Embeddium renderer), render distance 8. Use if Acedium misbehaves.
- **Distant Horizons** — optional, off by default (VRAM-heavy on a mobile-class card).

13. Risks, conflicts & mitigations

Risk	Severity	Mitigation
ParCool ↔ Player Animator clashes	Medium	ParCool+ Compatibility++ exists for exactly this
Acedium Sodiumized is an unofficial fork; disables under shaders	Medium	Toggable <code>compat</code> profile; auto-fallback to stock Embeddium
Backported Spears is a community backport (public domain, well-downloaded)	Low–Med	Config toggles for lunge-in-water/gliding and hunger cutoff; alternative: Spear Core (API + spear system)
Attribute-swap automation needs a custom mod (no NeoForge port of ghostswap)	Low	Manual MC-28289 technique always works; we build the automation as a small mod
Millénaire 1.21.1 port is community/alpha	High	D7 decision; fallback = vanilla villages + Towns and Towers + KubeJS factions
Alex's Caves has no official 1.21.1 port	High	Isolate; make Chapter 6 skippable
Alex's Mobs has no official 1.21.1 port	Medium	Use Continued/unofficial port; pin build
Terralith + many structure mods = worldgen cost on infinite worlds	Medium	Density tuning; Tectonic off by default (D6)
Vanilla combat may feel thin vs. modded bosses	Medium	Boss design must respect vanilla mechanics (knockback, shields, mace Breach, bow charge); spear charge + parkour are the answer to "how do I deal damage safely"
Balance is guesswork until played	High	M7 is a real playthrough pass

60–100h may be too long	Low	Ship Act I (Ch 0–5) first
SWEM replaces the vanilla horse model — clashes with any other entity-model mod	Medium	No other entity-replacing mods in the stack; fallback = vanilla horses
Sophisticated Backpacks + Curios — backpack-as-curios slot interactions	Low–Med	Test in M3b; disable Curios integration if it misbehaves
FallingTree + Terralith's giant trees — one-cut felling of a 40-block tree can trivialise wood	Medium	Woodcutting-level cap on logs felled per cut; shift-click bypass for safety
Supplementaries is content-heavy (11+ MB, hundreds of blocks)	Medium	Cut first per §6.11 budget rule
Guard Villagers + Millénaire — guard spawns inside modded villages	Medium	Config guard counts; verify Millénaire villages aren't over-run
Dynamic Waters + Flowing Fluids both modify water — potential conflict/duplication	Medium	Test together in M3b; if they fight, keep Dynamic Waters (worldgen rivers) and drop Flowing Fluids, or vice versa
Dynamic Waters requires a brand-new world	Low (we are building a new world)	Lock the water stack <i>before</i> M3 worldgen; never add it to an existing save

River mods conflict with mods that modify vanilla rivers	Low–Med	Do not add vanilla-river-overriding mods alongside Dynamic Waters/Streams Reflowing
Sable/Sablewaves is a physics engine — CPU cost, newer/less proven mod	High	Perf-gated and optional; cut first per §6.11 budget rule
YUNG's Better Mineshafts vs Hopo Better Mineshaft — both replace mineshafts	Medium	Use YUNG's only; do not install Hopo's mineshaft variant
Repurposed Structures flood — adds many variants; spawn rates must be tuned	Medium	Configure per-structure rates and dimension/biome blacklists before worldgen
Structure-mod density raises chest/spawner counts — CPU/entity cost	Medium	Entity caps + density tuning; verify tick time in M9
The Graveyard's lich chain needs villager kills (Vial of Blood) — conflicts with the standing system	Medium	Rework the ritual in KubeJS to use a non-villager source, or gate the quest behind nations that tolerate it
Many structure mods slow Explorer's Compass searches	Low	Compass handles the registry fine; blacklist noise structures
Fluid tick cost at scale (oceans, dams, large water volumes)	Medium	Configure Flowing Fluids' performance settings; cap water-heavy regions; watch tick time in M9

Iron's Spells adds 18+ attributes to the pool Attribute Modify and Reforged also write to	Medium	Attribute Modify touches <i>base</i> attributes only and leaves the <code>irons_spellbooks</code> namespace alone; test in M5
Wizard NPCs selling scrolls breaks rule 9 (traders sell no power)	Low	Strip the scroll trades in KubeJS — they sell materials only
Cast times vs. parkour — a frozen cast input can feel wrong next to ParCool	Medium	Decide in M1 (default: rolling wastes the cast, it is not cancelled); test feel early
Monsters & Spellbooks addon is under active development and likely to be reworked	Medium	Optional; cut it before touching anything in the core stack

14. Build plan

Milestone	Deliverable	Effort
M0 — Skeleton	manifest, mod folder, boot test, perf baseline	1 session
M1 — Movement & combat	Backported Spears, ParCool, Combat Roll, Compatibility++, GhostSwap mod, tuning	1 session
M2 — Soulslike layer	Silent's Power Scale, GraveStone, Ember Flask prototype	1 session
M3 — Worldgen	Terralith + Explorify + D&T + T&T + WDA, density tuning, starter-region pregen	2 sessions

M3c — Dungeon layer	YUNG's suite + Repurposed Structures + Hopo + The Graveyard + Structory + Dimensional Dungeons; structure density/ spawn-rate tuning; loot-table injection for Codex pages and unique loot	2 sessions
M3b — World fullness & water	Lock the water stack first (new-world requirement) , then FallingTree (Woodcutting gate), Sophisticated Backpacks (+upgrades), SWEM (nation horse lines), Guard Villagers (raid→standing), Farmer's Delight, Dynamic Waters + Flowing Fluids + Boatload, integration tuning	2 sessions
M4 — Nations	Millénaire decision (AW2 is out), standing system in KubeJS	1–2 sessions
M5 — RPG & grinding	Project MMO + Classes + Origins + skill books + Mob Grinding Utils + Reforged + Attribute Modify	1 session
M6 — Boss spine	Cataclysm + BOMD + Aether + Hollow Court integration, boss order test	2 sessions
M7 — Landmarks, puzzles & story	Landmark structures, Annals, puzzle logic, journal, FTB Quests chapters 0–8	3 sessions
M8 — Crafting & loot	KubeJS gating, signature weapons, nation ultimates, Reforged tuning	2 sessions
M9 — Balance & perf	Full playthrough pass, FPS/RAM verification on your GPU	2 sessions

15. Open decisions

ID	Decision	My recommendation
D1	Combat engine	Vanilla + ParCool + Combat Roll + spear ✓ settled
D2	Scope for first buildable version	Act I (Ch 0–5) first, then extend
D3	✓ RESOLVED — magic is core. Iron's Spells 'n Spellbooks is in from the start; the Hexer class is live	
D4	Death severity	Ember loss + Hollowing capped at –20%
D5	Alex's Caves (unofficial port)	Include, make Chapter 6 skippable
D6	Tectonic (dramatic continents, heavier worldgen) on an infinite world	Off by default — Terralith + Explorify carry it; Tectonic as an optional add-on profile
D7	Nations: Millénaire (rich, alpha, ~1.2k downloads on 9.0.2) vs vanilla + Towns & Towers + KubeJS-only factions	Millénaire in — AW2 is <i>out</i> (no 1.21.1 build), so this is now only a Millénaire in/out call. Decide after a boot test; the fallback is fully specified
D8	Starter pregen radius	~10k × 10k
D9	Expert recipe rebalance	Full — core to "difficult"
D10	Fast travel	Waystones unlocked per chapter
D11	NG+ / New Pilgrimage	Design the hook now, implement later
D12	Acedium default	On , with <code>compat</code> profile as fallback

D13	Distant Horizons	Off by default; optional profile
D14	✓ RESOLVED — the RECOMMENDATIONS T1 batch (boss bars, damage indicators, loot beams, Mob Champions, boss music, Traveler's Titles, runes, Threat tiers) is approved and in the stack	
D15	✓ RESOLVED — Monsters & Spellbooks (Necro school) is in , not optional	
D16	✓ RESOLVED — the Witchbane standing penalty for a Hexer stays at -40	

16. Sources (verified 1.21.1 NeoForge unless noted)

Spear / combat / movement

1. Minecraft 1.21.11 "Mounts of Mayhem" — spear mechanics (jab, charge, speed-scaled damage, Lunge, `kinetic_weapon` component, Mob Kabob): <https://www.akliz.net/blog/posts/minecraft-12111-mounts-of-mayhem> · <https://www.minecraftforge.net/news/60948-java-edition-1-21-11> · https://minecraft.fandom.com/wiki/Mounts_of_Mayhem
2. Backported Spears (NeoForge 1.21.1, 1.8.0 — jab, charge, Lunge, config toggles): <https://www.curseforge.com/minecraft/mc-mods/backported-spears/files/all> · <https://modrinth.com/mod/backported-spears> · <https://www.planetminecraft.com/mod/backported-spears/>
3. Spear Core (alternative spear API for 1.21.1 NeoForge): <https://www.modpackindex.com/mod/150879/spear-core>

4. Combat Roll 2.0.6+1.21.1: <https://www.curseforge.com/minecraft/mc-mods/combat-roll/files/7348018>
5. ParCool (NeoForge 1.21.1): <https://www.feed-the-mods.com/parcool-mod/> · ParCool+ Compatibility++ (ParCool ↔ Player Animator ↔ Better Combat): <https://www.curseforge.com/minecraft/mc-mods/parcool-compatibility-addon-neoforge-edition>
6. Attribute swapping — MC-28289 behaviour, Mojang classifies as exploit: https://www.reddit.com/r/admincraft/comments/1rs5tj7/attribute_swapping_not_working_12111/ · Fabric-only automation mods (no NeoForge port): <https://www.curseforge.com/minecraft/mc-mods/auto-attribute-swap>
7. Attribute Modify (NeoForge 1.21.1, datapack-driven attribute tuning incl. Curios): <https://www.curseforge.com/minecraft/mc-mods/attribute-modify> · <https://modrinth.com/mod/attribute-modify>
8. Reforged (Tiered fork — gear tiers/modifiers + stat rerolling): <https://modrinth.com/mod/tiered>

Bosses / dimensions 9. L_Ender's Cataclysm: <https://minecrafthub.io/mods/l-enders-cataclysm> · <https://www.feed-the-mods.com/l-enders-cataclysm-mod/> 10. Bosses of Mass Destruction: <https://www.curseforge.com/minecraft/mc-mods/bosses-of-mass-destruction-forge> 11. The Twilight Forest 1.21.1: <https://www.curseforge.com/minecraft/mc-mods/the-twilight-forest> 12. The Aether 1.21.1: <https://modrinth.com/mod/aether/version/1.21.1-1.5.6-neoforge> 13. Deeper and Darker: <https://www.curseforge.com/minecraft/mc-mods/deeperdarker> 14. Alex's Caves (Unofficial Port 2.0.9): [https://modrinth.com/mod/alexs-caves-\(unofficial-port\)/version/2.0.9](https://modrinth.com/mod/alexs-caves-(unofficial-port)/version/2.0.9) 15. Alex's Mobs 1.21.1 ports: <https://www.curseforge.com/minecraft/mc-mods/alexs-mobs-1-21-1-port/files/all> · <https://modrinth.com/project/kYlaHHfw> 16. Iron's Spells 'n Spellbooks 1.21.1: <https://modrinth.com/mod/irons-spells-n-spellbooks/version/1.21.1-3.11.0>

Nations / worldgen / discovery 17. Millénaire 1.21.1 NeoForge (7 cultures, reputation, diplomacy, quests, chronicles): <https://www.millenaire.org/> · <https://github.com/Leviaria/Millenaire> · <https://github.com/gblfxt/Millenaire-rewrite-1.21.1> 18. ~~Ancient Warfare 2~~ — **no 1.21.1 build**. CurseForge project 224602 stops at 1.12.2 (last update 2021), so it is out of the pack. Recorded in [dropped] in tools/modlist.toml. 19. Terralith: <https://www.curseforge.com/minecraft/mc-mods/terralith/files/all> · Tectonic: <https://modrinth.com/datapack/>

tectonic 20. Explorify (1.21.1, vanilla-friendly structures): <https://www.curseforge.com/minecraft/mc-mods/explorify> 21. Dungeons and Taverns: <https://minecrafthub.io/datapacks/dungeons-and-taverns> · When Dungeons Arise: <https://minecrafthub.io/mods/when-dungeons-arise> 22. Towns and Towers: <https://www.curseforge.com/minecraft/mc-mods/towns-and-towers> 23. Explorer's Compass (structure locator, finds any mod's structures): <https://www.curseforge.com/minecraft/mc-mods/explorers-compass> · Nature's Compass: <https://www.curseforge.com/minecraft/mc-mods/natures-compass> 24. YUNG's Better Dungeons: <https://www.curseforge.com/minecraft/mc-mods/yungs-better-dungeons-neoforge>

RPG / loot / quests / difficulty 25. Project MMO + Skill Books + XP Bottles: <https://modrinth.com/mod/project-mmo-skill-books> · <https://www.curseforge.com/minecraft/mc-mods/project-mmo-xp-bottles> 26. Mob Grinding Utils 1.21.1: <https://www.curseforge.com/minecraft/mc-mods/mob-grinding-utils> 27. FTB Quests: <https://www.curseforge.com/minecraft/mc-mods/ftb-quests-forge> 28. Lootr: <https://modrinth.com/mod/lootr/version/1.21.1-1.11.36.116> 29. Artifacts: <https://www.curseforge.com/minecraft/mc-mods/artifacts> · Enigmatic Legacy+: <https://modrinth.com/mod/enigmatic-legacy-+-retrieve-to-slots> 30. Simply Swords: <https://modrinth.com/mod/simple-swords/version/OpiTyBBv> · Spartan Weaponry: <https://www.curseforge.com/minecraft/search?class=mc-mods&search=Spartan+Weaponry> 31. Origins: Classes: <https://www.curseforge.com/minecraft/mc-mods/origins-classes-neoforge> 32. **Silent's Power Scale** (the successor to Scaling Health — same author, scaling-attribute system for players and mobs): <https://modrinth.com/mod/silents-power-scale> · source: <https://github.com/SilentChaos512/Power-Scale>. *Scaling Health itself has no 1.21.1 build (CurseForge project 248027 stops at 1.20.1); Customizable Mob Difficulty Scaling has no 1.21.1 NeoForge build at all.* 33. GraveStone Mod: <https://www.curseforge.com/minecraft/mc-mods/gravestone-mod> 34. Waystones: <https://minecrafthub.io/mods/waystones> · Fast Travel Waypoints: <https://www.curseforge.com/minecraft/mc-mods/fast-travel-waypoints> 35. Curios API: <https://modrinth.com/mod/curios/version/9.4.2+1.21.1> · Lionfish API: <https://www.curseforge.com/minecraft/mc-mods/lionfish-api> 36. Soulslike Deaths (optional): <https://modrinth.com/mod/soulslike-deaths>

Performance 37. Acedium Sodiumized (NeoForge 1.21.1 Nvidia port;

requires NVIDIA GTX 1600+/RTX 20xx+): <https://www.curseforge.com/minecraft/mc-mods/acedium-sodiumized> · <https://github.com/viktorr245/acedium-sodiumized> 38. NeoForge 1.21.1 optimization mod recommendations: https://www.reddit.com/r/feedthebeast/comments/1rss3o1/what_performance_mods_neoforge_1211_are_actually/ · <https://champbop.com/minecraft/best-optimization-mods-minecraft/>

Structures & dungeons 52. YUNG's Better suite, all on 1.21.1 NeoForge (Dungeons, Strongholds, Mineshafts, Ocean Monuments, Nether Fortresses, Desert Temples, Jungle Temples, Witch Huts, End Island, Bridges, Extras, API): https://www.reddit.com/r/feedthebeast/comments/1q2f5f3/all_yungs_better_mods_are_now_available_on_1211/ · <https://www.curseforge.com/minecraft/mc-mods/yungs-better-dungeons-neoforge> · <https://www.curseforge.com/minecraft/mc-mods/yungs-better-nether-fortresses-neoforge> 53. Repurposed Structures (NeoForge 1.21.1 — biome variants of vanilla structures; Jungle Fortress, distant End strongholds; serverside, vanilla-client-safe): <https://modrinth.com/project/QDNS5oAT> · <https://www.curseforge.com/minecraft/mc-mods/repurposed-structures> 54. The Graveyard (FORGE/NEOFORGE) 1.21.1 NeoForge — 10+ lore structures, Lich Prison, the Lich's Bargain quest chain, bone staff fragments, Vial of Blood ritual: <https://www.curseforge.com/minecraft/mc-mods/the-graveyard-forge> · The Graveyard - Resurrected (1.21.1 NeoForge): <https://modrinth.com/mod/graveyard> 55. Structory (NeoForge 1.21.1 — ruins, towers, cottages, stables, graveyards, settlements, boats; Terralith-compatible): <https://www.curseforge.com/minecraft/mc-mods/structory> 56. Hopo Better Ruined Portals / Hopo Better Mineshaft / Hopo Better Underwater Ruins (NeoForge 1.21.1): <https://www.curseforge.com/minecraft/mc-mods/hopo-better-ruined-portals> · Loot Integrations addon for Hopo structures: <https://www.curseforge.com/minecraft/mc-mods/loot-integrations-hopo-better-mineshaft-ruins> 57. Dimensional Dungeons (NeoForge 1.21.1 v210 — renewable generated dungeon dimension, Patchouli guidebook): <https://www.curseforge.com/minecraft/mc-mods/dimensional-dungeons>

Water & sailing 46. Dynamic Waters: Realistic Flowing Rivers (NeoForge 1.21.1 — worldgen-carved rivers, downhill flow, boat/item currents; Terralith-compatible; new world required): <https://>

www.curseforge.com/minecraft/mc-mods/dynamic-waters 47. Streams Reflowing (NeoForge 1.21.1 — alternative river/current mod with vessel drift settings): <https://modrinth.com/mod/streams-reflowing> 48. Flowing Fluids (NeoForge 1.21.1, by Traben — realistic fluid physics, pressure system, per-entity push control): <https://www.curseforge.com/minecraft/mc-mods/flowing-fluids> 49. Sablewaves (NeoForge 1.21.1 — moving sea surface, ocean currents, moon tides, storm swells, wind; requires Sable 2.0+): <https://www.curseforge.com/minecraft/mc-mods/sablewaves> 50. Currents of Trade (NeoForge 1.21.1 — harbour network, sailing sloops, cargo fleets between coastal villages): <https://github.com/Lexovian/Currents-of-Trade> 51. Boatload (NeoForge 1.21.1 — alternative sailing vessels): <https://www.curseforge.com/minecraft/search?class=mc-mods&search=boat>

World fullness 40. FallingTree (NeoForge 1.21.1 — modes, whole-tree detection, tag support): <https://minecrafthub.io/mods/falling-tree> · Panda's Falling Trees (animated topple alternative): <https://www.feed-the-mods.com/pandas-falling-trees-mod/> 41. Sophisticated Backpacks 1.21.1 NeoForge (upgrades incl. tool swapper, tank, mob catcher; requires Sophisticated Core): <https://modrinth.com/mod/sophisticated-backpacks/version/1.21.1-3.25.51.1801> 42. Star Worm Equestrian (SWEM) 1.21.1 NeoForge (gaits, tack, bonding, genetics, racks): <https://www.curseforge.com/minecraft/mc-mods/swem> 43. Guard Villagers 1.21.1 NeoForge (village guards, configurable, cleric healing, golem repair): <https://www.curseforge.com/minecraft/mc-mods/guard-villagers> 44. Farmer's Delight 1.21.1 NeoForge (crops, rich soil, cutting board, stove cooking): <https://minecrafthub.io/mods/farmers-delight> 45. Supplementaries 1.21.1 NeoForge (vanilla+ blocks/items): <https://www.curseforge.com/minecraft/mc-mods/supplementaries>

Inspiration 39. The Ultimate Survival World (RareLoot / TrixyBlox & StephBlox — island world, transformed vanilla biomes & structures, peculiar secrets, epic lore, tricky challenges, 12+ biomes, 2 dimensions): <https://www.bedrockexplorer.com/@rareloot/the-ultimate-survival-world> · <https://www.sportskeeda.com/minecraft/how-play-minecraft-ultimate-survival-world> · <https://nullforums.net/resources/the-ultimate-survival-world-by-rareloot.12356/> · https://ultimate-survival-world.fandom.com/wiki/Ultimate_Survival_World_Wiki

*End of design document v0.7. Nothing built yet — this is the blueprint for review. Next: your approval, then M0 (the pack skeleton). built yet — this is the blueprint for review.**